

Evaluation & Study Design

Day 3 — How would you know if it worked?

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Where we are on the spine

Handoff — participants recap their Day 2 data specification.

- **Yesterday:** a problem → the data that exists to attack it.
- **Today:** data → a **credible way to know** whether a solution works.

🔗 Day 2: provenance & who is missing

The central question

Before you build anything:

What would count as success?

A method you **cannot evaluate credibly** is not yet a contribution.

The validity ladder

"Evaluation" otherwise smears across three days. Name the rungs:

- **Internal validity** — is the model sound on its own terms? (*Days 3–4*)
- **External / deployment validity** — does it hold in the world? (*Day 5*)

Today is the bottom rung: study design.

FIG F3.1 The validity ladder. A two/three-rung ladder: study design → internal validity (Days 3–4) → external/deployment validity (Day 5). Reused on Day 4 and Day 5 to show the rung change. *See build guide F3.1.*

Two things both called "evaluation"

- **(a) Study design** — defining the *target* and how you would credibly *measure* it. ← **today.**
- **(b) Evaluating a model you have built** — Days 4 and 5.

Today is **(a)**: you design the evaluation **before the model exists.**

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NAMING THE TARGET

Defining what you are actually measuring

Estimands — what are you actually measuring?

- Name the **quantity of interest** precisely — *before* choosing a metric.
- A vague target produces an **unfalsifiable claim**: anything can look like success.

Vague (not an estimand)

"Improve sepsis prediction."

Precise (an estimand)

"Sensitivity at a 90%-specificity threshold, in adult ICU admissions, for sepsis onset within 6 hours."

The anatomy of an estimand

Every credible target names four things:

ELEMENT	THE SEPSIS EXAMPLE
Population	adult ICU admissions
Outcome	sepsis onset
Timing / horizon	within 6 hours
Operating point	sensitivity at 90% specificity

Write these down before you touch a model. If you can't, you don't yet have an evaluable claim.

Target population vs. deployment population

- The population you **evaluate on** is rarely the population you **deploy on**.
- The gap is a **study-design decision**, not an afterthought.
- Who is in the evaluation set is inherited from the data's provenance.

FIG F3.3 Eval vs. deploy population. Two partly-overlapping distributions (e.g., age or acuity) labeled "evaluation cohort" and "deployment population," with the non-overlap shaded as the generalization gap. *See build guide F3.3.*

🔗 Day 2: who is missing

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CHOOSING WHAT TO MEASURE

Metrics

Metrics: utility vs. convenience

- **Metrics of convenience** — AUROC for everything, because it is easy.
- **Metrics of utility** — chosen because they track the **clinical decision**.

TOOL	TELLS YOU	DOES <i>NOT</i> TELL YOU
Discrimination (AUROC)	Ranking ability	Whether probabilities are usable
Calibration	Are the probabilities honest	How well it ranks
Decision curves	Net benefit at real thresholds	Anything outside that range

Choose the metric the clinical decision implies.

Discrimination — and its limits

- **AUROC** — probability a random positive outranks a random negative; prevalence-independent.
- **AUPRC** — area under the precision–recall curve; prevalence-dependent.
- Both are **threshold-free summaries**: they rank models but say nothing about whether the *probabilities* are usable, or whether any single operating point helps a decision.

Discrimination is necessary, not sufficient. Calibration and decision utility come next.

At the operating point, prevalence still bites

A sepsis screen, **1% prevalence**, AUROC = 0.90, threshold set for **80% sensitivity**:

- At a plausible specificity, **most positive alarms are false** — PPV can sit in the single digits.
- The clinician experiences **alarm fatigue**, not a 0.90 model.
- A strong threshold-free summary can coexist with a useless **operating point**.

→ Day 5: alarm fatigue

Calibration

- A well-discriminating model can be **badly miscalibrated**.
- The moment a prediction **informs a decision**, calibration is what matters.
- "70%" must mean 70% — or the threshold you chose means nothing.

FIG F3.5 Reliability curve. Predicted probability (x) vs. observed frequency (y), the diagonal "perfect calibration" line, and a miscalibrated model's curve deviating from it. *See build guide F3.5.*

Clinical utility — net benefit

- **Decision-curve analysis** — does using the model beat "treat all" / "treat none"?
- Choose the **operating threshold from the actual decision**, not the metric.
- Net benefit folds in the **relative cost** of false positives and false negatives.

FIG F3.6 Decision curve. Net benefit (y) vs. threshold probability (x), with the model curve above "treat all" and "treat none" reference lines over the clinically relevant threshold range. *See build guide F3.6.*

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SUBGROUPS, POWER, AND VALIDITY THREATS

What aggregate numbers hide



Fairness and disaggregated evaluation

- **Aggregate performance can hide subgroup failure** — a strong overall number can mask a model that fails an entire population.
- **Disaggregated evaluation** is a default, not an add-on.
- Decide the subgroups, and the per-subgroup targets, *as part of the design*.

FIG F3.7 Aggregate hides subgroups. One reassuring overall bar beside a panel of per-subgroup bars, one of which is clearly failing. See *build guide F3.7*.

Statistical power

- **Underpowered studies produce unreliable claims** — including unreliable *positive* ones (Card et al., "With Little Power").
- Power is a **design decision**, made now — not an afterthought at review time.
- Ask early: how much data, how many events, to detect an effect that would matter?

The medical algorithmic audit

A **structured framework** for evaluating a clinical model end to end (Liu et al., 2022) — a named thing participants can apply.

- Specify the intended use, then probe systematically for failure.
- Test subgroups, edge cases, and the points where the model meets the workflow.
- An auditable model is one whose failures **don't hide**.

Evidentiary standards — the bar for a real claim

- **RCTs of ML interventions** (Plana et al., 2022) — the high bar for a *deployed* claim; few clear it.
- **Effect heterogeneity** (Yu et al., 2024) — an average effect can hide who is helped and who is harmed.
- "Works on average" ≠ "shown to help patients."

Threats to internal validity

The quiet study-killers — name them in advance:

- **Leakage** — information from the future, or from the label, in the features.
- **Shortcut learning** — the model solves a proxy, not the task.
- **Development-to-deployment shift** — the eval set isn't the deployment world.

These invalidate a study *silently*.

Worked example — leakage

How a beautiful AUROC is manufactured by accident:

- A feature is only recorded **after** the outcome is known (e.g., a discharge code, a treatment that follows diagnosis).
- The model "predicts" the outcome by reading its **consequence**.
- It collapses the moment that feature isn't available **before** the decision.

Leakage is the most common and most embarrassing failure. Hunt for it.

The afternoon — build your own evaluation

The 2:45 notebook, end to end, on your project:

- 1 **Define the cohort** — your estimand, made executable.
- 2 **Discrimination + calibration** — AUROC/AUPRC and a reliability curve.
- 3 **Disaggregated audit** — per-subgroup performance.
- 4 **Decision-curve analysis** — net benefit at your real threshold.

Worked example, template, and AI-coding-tool guidance are provided — see the Day 3 & 4 Workshops doc.

Dissect, then design

Today's deliverable — Evaluation & study-design plan.

Your **estimand**, the **metrics** the clinical decision implies, your **subgroup/fairness** evaluation, and the top **threats to validity** for your project.

Now (9:00): in pairs — dissect a published evaluation. *Does this evidence convince you?*

Next: 11:00 guest (Olawale Salaudeen) — disaggregated metrics · 1:30 apply the critique lens · 2:45 evaluation notebook.